

Exploring Brain-Computer Interfaces in Gaming - A Short Review

Faris Abuhashish

Computer Science - VR Department

University of Petra

Amman, Jordan

faris.abuhashish@uop.edu.jo

Anas W. Arram

Department of Computer Information Science

Higher Colleges of Technology

Dubai, UAE

aarram@hct.ac.ae

Abstract—Due to the advanced technologies in gaming, new interfaces were created and the usage of brain-computer interaction or BCI. BCIs use neural signals to act as interfaces that control signal interaction between the brain and external devices, and as such can be seen as substitutes or additions to game controllers. Examining the prospect of applying this technology in the context of its target market; video games, its components, its modes of signal acquisition/processing, the present envision BCI as the spearhead of a new revolution in gaming. As the most common and non-invasive and relatively cheap technique, EEG forms the basis of measuring brain activity while other methods are also covered. The paper assesses two machine learning approaches in decoding neural signals and reviews the diverse types of video games that may include BCI interfaces. In addition, the paper presents issues: limitations of hardware and software, variation in subjects' performances and lack of unity, and lastly, perspectives of BCI games for therapeutic and educational purposes. To this end, this paper seeks to foresee potential solutions to the challenges discussed above so as to help lead further innovation in the field. The presented research provides evidence of BCI as an innovative solution for gaming by integrating entertainment, neuroscience, and technology to enhance the concept of gaming experience.

Index Terms—Brain-Computer Interfaces, BCI in Gaming, EEG Signal Processing, Machine Learning in BCIs, Gaming Technologies.

I. INTRODUCTION

Computer conventional games have conventionally employed conventional human interface apparatus in form of keyboard, mouse, gamepad, or special controllers in interacting with virtual realities [1]. Such tools are meant to map a player's real-world movements onto actions within the game or choices to provide control and accuracy. However, they also have some drawbacks, namely, the primary reliance on the user's fine motor skills and thus, the impossibility of use for physically disabled persons. The more advanced gaming is today, people start to think about a need for interfaces that are free from such limitations.

Brain-Computer Interaction (BCI) means a new unprecedented advancement in gaming technology where players control the game with their brains [2]. Basing on the same conception, BCIs establish direct link between the person and the game as it defines the electrical activity in the brain [3]. This revolutionary concept avoids motor activity, enabling people to control games only with their minds or, to be

more specific, intellect. It makes technologies that reveal new perspectives and possibilities of approach to playing games in perspective, immersed and adaptive forms.

In addition to acting as alternatives to the classical controls, BCIs can enhance the existing input by including cognitive and emotional feed-back [3]. For instance, BCIs can challenge the user at different levels depending on his or her psychological condition or present feedback in the course of the game [3]. This duality improves gaming and results in the primary mode of use and opens new opportunities for use in fun and therapy.

Nevertheless, the integration of BCI technology in gameplay has only started and still has multiple constraints which must be investigated among which we have the limits of the hardware, the variability of signals and adaptation of the user [4]. This paper reviews the current trend of BCI in gaming, defines basic principles, and concepts and points out the challenges that must be met to transform interfaces to gaming for a connected & inclusive future. Figure 1 illustrates playing a game using BCI.

This review specifically focuses on EEG-based VR systems particularly the DSI-VR300 along with their use in gaming while explaining how these setup methods enhance user immersion and stabilize signals. Our proposed framework connects traditional EEG assessment to VR methods to create an unexplored BCI gaming solution according to previous research findings.

This review paper is ordered as follows: Section II delivers background on BCI technologies; Section III deliberates methods for measuring neural activity; meanwhile section IV reviews BCI input methods; Section V summaries EEG signal preprocessing and feature extraction; Section VI delegates machine learning techniques for EEG classification, together with the novel argument on RNN architectures; Section VII inspects BCI apps in video games with a emphasis on VR integration; Section VIII deliberates limitations and proposes directions for future research; eventually, Section IX accomplishes the paper.

II. BACKGROUND OF THE STUDY

Brain-Computer Interfaces (BCIs) are novel interfaces that link the user brains with external devices by analyzing neural signals [5]–[7]. What is more, using the approach that minimizes the interference of other input factors, BCIs allow



Fig. 1. Playing game using BCI.

users control the targeted applications merely by their brain signals [8], [9]. One way of documenting this activity is Electroencephalogram (EEG), a Brain Mapping technology free from invasive tools, which picks brain's electrical output. Such oscillatory signals, also known as brain waves, are generated through neuronal electrochemical processes and travel from the cerebral cortex to the scalp [10]–[12]. EEG equipment records this activity with electrodes located on the head, without the need for surgery. The accumulated signals are next gained, filtered and readied for use in a BCI that decodes it into proper signals within target applications [13]–[15].

BCI systems principally rely on a procedure referred to as self-regulation whereby a user consciously changes his or her brain activity in order to communicate with the system. Neurofeedback—providing users with information about specific electrical activity within the brain—appears to be important in self-regulatory mechanisms [16]–[18]. In this closed-loop neurofeedback system, the EEG signal is acquired, analyzed and converted into control signals for the application. The result is a procession of creation and feedback comments, which in turn to the users of the application. This feedback loop makes the next control because it allows users to work on their brain wave patterns to achieve a particular result.

An often-important aspect of BCI systems is the translation stage that utilizes methods, like classification or regression, in order to map filtered EEG features into control signals [19]–[21]. These control signals are distinguished into endogenous ones and exogenous ones. Intrinsic control signals arise out of the intra-user drive and are not a result of any exogenous cue. Some of them include motor imagery where the users imagine moving body parts and body signals or measurements of the user's cognitive or emotional state [22]–[24]. However,

exogenous control signals are evoked by visual or other peripheral cues such as flashing lights, which are popular in EEG-based BCI research.

Common exogenous signals include the P300 event-related potential and the Steady-State Visually Evoked Potentials (SSVEPs) [25]–[27]. The P300 is a voltage change elicited through an external input with the intention of obtaining a response, whereas SSVEP is an oscillatory response to visual stimuli which flicker at predefined frequencies. Each signal type has been investigated for its stability in providing meaningful commands in BCI systems. These differentiated endogenous and exogenous signals are meant to capture the versatility of BCIs which are seen as a promising technology for various uses in practice in the domains of gaming, rehabilitation, etc. Figure 2 illustrates a flow diagram while playing games using BCI.

In addition to the qualitative review provided in this paper, future research should incorporate quantitative evaluations. A proposed framework includes controlled experiments using standardized VR-BCI setups to measure key performance metrics such as cross-validation accuracy and processing latency. This would allow for a more rigorous comparison of various BCI systems in gaming applications.

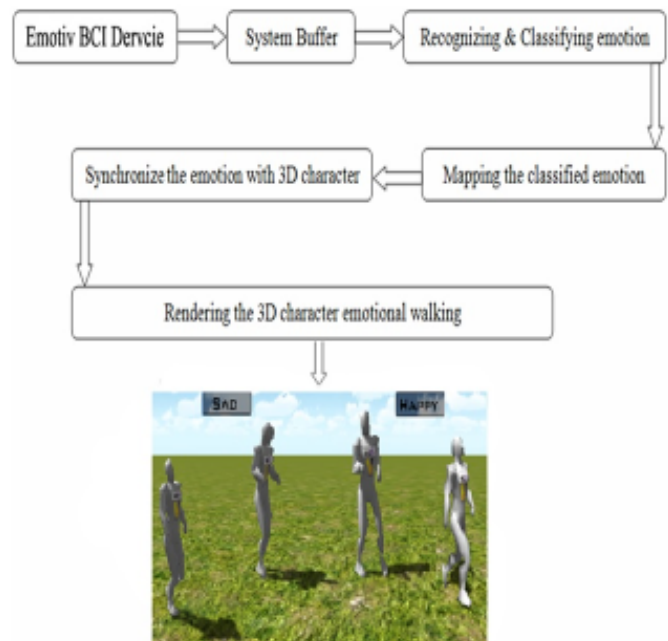


Fig. 2. Flow diagram while playing games using BCI.

III. MEASURING NEURAL ACTIVITY

One of the issues of any BCI system is the choice of an appropriate method for the monitoring and recording of neural signals. BCI creates a connection between the human brain and an external device without engaging pathways as the peripheral nervous system. This provides immediate, cyclic feedback between the brain input and the system output as well as the user's intention. The central step in it is the recordation

of the electricity glazing in the brain, which is very weak but registrable by equipment. These signals, mainly recorded from the head region of brains, can be recorded with the help of non-invasive scalp electrodes or by implanting electrodes in the brain.

Electrical techniques including implantation of micro-electrodes on cortical or subcortical areas provide temporal issue which enables determination of activity of single neurons. However, these techniques are not without their problems; one is the risks associated with surgery, second is likelihood of neural irritation, and third is the time factor in that the electrodes can wear out. Other techniques, for example, electroencephalography (EEG), are more secure and easily available. EEG uses electrodes on the scalp to capture electrical activity passing through the skull and is cheaper to perform and carries no risks of surgery. However, the proposed method is less temporal accurate than invasive techniques and is sensitive to noise.

Generally, in EEG systems, reference electrodes are applied to record brain activity in two zones and compare these records by working with a stable zone out of the head such as ears, or sagittal plane of head. Extensor electrodes are placed corresponding to 10-20 international system which indicates the exact margins of the brain. For instance, frontal and face-basing areas are abbreviated as F, temporal and ear-basing areas as T, central and shoulder-basing areas as C, parietal and vertex-basing areas as P, occipital and crown-basing areas as O, according to the complexity of test required, either with fifteen electrodes system or extended ten-twenty system is used.

EEG signals are divided by the bands like delta, theta, alpha, beta and gamma bands; delta band ranges from 1 Hz through 4 Hz; theta band ranges from 4 Hz through 7 Hz; alpha band ranges from 8 Hz through 13 Hz; beta band ranges from 13 Hz through 30 Hz; and last is gamma band; greater than 30 Hz. These frequencies are made up of the time, frequency and spatial domain which act as categories that sort raw data for purposes of analysis. Sometimes, extra peripherals such as electromyograms (EMG) and galvanic skin response (GSR) are included in order to gain better and more accurate signal classification. These PNS tools enhance EEG data yielding a high BCIs' accuracy in various applications like gaming, rehabilitation, or research.

IV. BCI INPUT METHODS

BCIs use numerous inputs acquired from various areas of the head; that in order to identify the activity, frequencies are used. There is, however, one kind of input that has been identified and its name is motor imagery (MI), which is derived from the sensorimotor area of the brain. Again, EMG accelerations are affected by movement, and thus MI is achieved by visualizing the movement of some body parts without actually performing the movement. This technique acts mainly on the mu and beta rhythm bands and is preferentially related to different cortical homunculus sectors, including hands, tongue, and lips. MI has been successfully applied in, for example,

controlling a wheelchair in 2D space, as shown by [28], and is commonly used in clinical rehabilitation for motor-disabled persons.

For these reasons, MI is an endogenous input technique, but single-state visually evoked potentials (SSVEPs) are considered to be exogenous. SSVEPs are elicited when the user directs their attention to a visible light signal or a flashing light with a specific frequency. These inputs are used to control actions in a program, for example in a game as tilting or firing some objects. Hence effective, SSVEPs are dependent on external structures such as the monitor refresh rate and interference of harmonics. Nevertheless, these challenges are not peculiar to the SSVEP, and the method is rather solid to common artifacts such as eye blinks for signal classification.

The second input mechanism is P300 potential, which is a voltage shift caused by an external event and which is assessed 300-500ms after the stimulus. This method is widely discussed in such paradigms when an odd stimulus is placed among repetitive ones to produce a different reaction. The P300 response is more specific to processes of interest and can be split into P3a, which is linked to attention via novelty, and P3b, which is related to working memory updating. This technique provides highly accurate and efficient control signals for BCI use.

Both endogenous and exogenous input mechanisms demonstrate the modularity of the BCIs in acquiring and translating the neural signals to functioning of real-world significance. All these methods bring specific advantages and specific problems, which create the opportunity for further developments aimed at the improvement of user interaction and control in BCI systems.

V. PREPROCESSING IN EEG SIGNAL ANALYSIS

Some of the key tasks to be performed before recording EEG data are categorized under pre-processing with a view to qualify signals intended to be used for analysis. Pre-emphasis is normally used to invert the high frequencies that tend to be filtered away by most filters. For instance, while applying band-pass filters which respond to a given range of frequencies, out the desired range is rejected while in the range it allows to pass all frequencies without attenuation in the passband and totally rejects certain frequency in the case of notch filters used to eliminate some interferences such as the power line noise. As suggested in [29], initial baseline data were filtered by applying 3-70 Hz Butterworth band-pass filter, with 50 Hz notch filter when performing comparison of various processing techniques and minimizing feedback loops.

EEG data analysis is done in several general domains which include time domain, frequency domain and spatial domain. The time domain analysis is quite easy while use of technique like Fourier Transform helps in doing frequency domain analysis, similarly the spatial analysis is also possible by mapping-based tools. About these domains features defining is a significant process useful for data pre-processing before utilizing it with machine learning algorithms.

In this set up, it is important to note that any unrelated structural features including eye blinks or muscle movements proved to be a challenge while analyzing the EEG. Independent component analysis, or ICA indeed eliminates artifacts; in contrast to PCA while maintaining signal multidimensionality. ICA can be applied to biomedical signal processing such as EEG or used to extract signals such as P3a and P3b components.

Others are the CSP and some of its derivatives that are CSSSP and RCSP that enhance spatial filtering for the motor imagery tasks [30]. Building on SLD, source localization is advanced even further to separate classifiers' ability to distinguish motor tasks more effectively [31]. It should be noted that many algorithms are known, but each of them has different efficiency indicators depending on certain applications.

A. Feature Extraction Techniques in EEG Signal Processing

The conversion of raw EEG data into discriminative features for classification depends on the essential process of feature extraction. In the domain three widely adopted techniques exist: Common Spatial Patterns (CSP) together with Wavelet Transforms and Principal Component Analysis (PCA) bring valuable features to the extraction process from raw EEG data.

VI. MACHINE LEARNING IN BCI SIGNAL CLASSIFICATION

Following the extraction of features from the EEG data, automatic signal classification mapping techniques such as machine learning algorithms are applied to transform as well as produce more meaningful signals. One of the most used procedures is separation of data into two parts: training and test sets. There is a method such as k -fold cross-validation which involves the bridging of the data into k partitions [29], [32], [33]. In every cycle there is the utilization of $k - 1$ partitions in the process of training, the rest of the partition is adopted in testing. This process of cross-validating is repeated k number of times so that each partition is used in testing, and this helps in detection of problems such as selection bias and overfitting [34]–[36]. For instance, [8], [37], [38] used 10-fold cross-validation to assess the classification models in a prototyped BCI gaming system.

The Support Vector Machines (SVM) is an algorithm that is commonly utilized, which is typical in binary classification where the data is classified on a set of hyperplanes. SVMs find the largest margin between two classes so as to improve the model's accuracy of classification. Kernels like linear, radial slope improve adaptability for different shapes of data. SVMs are used in the BCI applications since they, as a class of ML algorithms, are accurate, robust, and fast as shown by [39].

Another widely used algorithm is Linear Discriminant Analysis (LDA), which is particularly used with problems in a low-dimensionality space. LDA works in a way of projecting data items to a one-dimensional line in order to maximize the distance between classes. In binary problems, this works well; however, in multi-class problems, one must use pairwise configurations or very sophisticated modifications, as discussed in [40].

Recent paradigms commonly used in BCI include deep learning methods such as convolutional neural networks (CNNs). CNNs transform raw EEG data or extracted features by convolutions and pooling in order to enhance significant features and minimize dimensionality. An FC layer completes this network and is used to classify the reduced data. CNNs are effective in dealing with large datasets; consequently, a simple Bayesian CNN was used by [41] in identifying P300 visual stimulus. This combination of new and original algorithms guarantees a stable picture in different BCI applications.

A. Benchmarking and Quantitative Evaluation

Standard benchmarks including cross-validation accuracy and real-time processing latency should be used by future empirical research to evaluate EEG-based BCI systems in gaming. We developed an assessment model to evaluate this system through experimentally controlled VR-BCI systems.

B. Recurrent Neural Networks for Time-Series EEG Analysis

The sequential nature of EEG signals makes Recurrent Neural Networks (RNNs) such as Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) suitable models for detecting temporal dependencies which Convolutional Neural Networks (CNNs) might miss. The signal classification process benefits from these models which either strengthen the performance of CNNs or outperform them particularly in situations requiring analysis of extended temporal patterns.

VII. BCI APPLICATIONS IN VIDEO GAMES

Several paradigms of BCIs have been successfully incorporated into many forms of in-game small-scale video games to check their viability. These include a pinball game as cited by [36], which also adopted Tetris for assessing the difference between P300 and MI paradigms and examined the aspect of cursor control, which underlies most game interfaces. [41] came up with MindGomoku, which is a Gomoku strategy game requiring Go pieces, and [42] developed a 3D simple coin collection game. [43] also discussed practical applications, including a virtual wheelchair control simulator.

Few studies have incorporated Virtual reality (VR) games, one of the immersive technologies, with BCIs. Although resulting in higher immersion, papers [44]–[47] demonstrate that classification accuracy is equivalent for abstract and realistic visual feedback. Challenges include costs of resources in VR game development and interference between EEG caps and VR headsets. Still, VR is applied in rehabilitation, as [48] have shown gratefully in their article.

BCIs have also been discussed in the context of complementary applications to gaming, such as using EEG signals as feedback for difficulty or interest rates. Chanel and colleagues tried to adapt game difficulty levels automatically based on EEG signals, with moderate classification performance. Similarly, [49], [50] employed EEG to measure user flow with faster training time but reasonable accuracy.

However, there are also other tools that have become very useful in integrating BCIs into games, for instance, OpenViBE

and OpenBCI. OpenViBE is an example of a user-friendly open-source software that can be adapted for use in multichannel EEG acquisition and analysis without the need for actual coding [51], [52]. Another implementation method for EEGs is interfacing the game engines with software that supports general scripting for the coordination of game design, such as Python or C++ [53].

Apart from entertainment, BCIs are also used in ‘serious games’ for treatment and training modes. For example, [54] demonstrated that FarmKeeper, a game based on BCI, increased attention in subjects with autism. A similar experiment for ADHD treatment with 2D and 3D BCI games was analyzed in [55], presenting distinguished experimental potential. Yet, relaxation or concentration enhancements were modest in the case of [56], and therefore further empirical studies should be encouraged regarding the efficiency of such games.

VIII. LIMITATIONS AND FURTHER WORK

However, there are many disadvantages of the BCI video games. A large part is hardware. The reality is that EEGs have reduced transfer rate as compared to the normal keyboard and mouse interface. They are also time-consuming in preparation, setting up and using in addition to a component of high cost [57]. The prices for consumer EEGs have come down, but it is not cheap enough to be regarded as a household article; [58] study identified that 16-electrode headsets can be bought for less than 300 Euros. [59] discusses the usability of a low-cost headset and reveals that after noise reduction the headset becomes quite possible, although there are some technical prejudices of this aid like no permission for medical use.

Almost all of the EEG headsets also use a wet applicative to improve electrical conductivity with the scalp. Dry processes have been studied [58] and concluded that it could be a potential replacement. It is reported that there are other materials that have been identified to display certain characteristics that could be as useful as wet electrodes and that dry electrodes are even more effective to be used when worn for longer hours. The overall advancement of the EEG is expanding; however, it is yet simple for more to be done before it can reach consumer level. Another possibility is the development of another efficient and convenient, that is, non-invasive method.

On the side of the subjects, issues are various performances using data within and between the subjects. BCI illiteracy, a concept where a person is unable to use a particular type of BCI such as MI, [60]. In fact, MI is potentially more discriminative than SSVEP or ERP when considering the occurrence of BCI illiteracy [56]. BCI illiteracy has also been considered to be a method that can, in fact, make researchers explain possible design of the experiment’s error to a static characteristic of a person. Another issue connected with subjects is that many experiments cause subjects to get bored while collecting data due to too much volume and repeatability of audiovisual material [61].

For the research subjects, the problem which exists today is that consistency and standardized procedures are not set and applied. As for neuroscience, more signals must be identified,

and more to the point, they must be easily separable and beneficial [62]. BCIs also have some ethical factors, which are psychological impact, practice, and policy making as proposed by [63]. Many of these studies are also lacking in video game design as well. A large reason for this is because research imposes that results must be forced quickly. Another part is the absence of cooperation with game developers and other cross-discipline specialists. Those coding these games are often lacking in game design knowledge with a larger skill in EEG signal analysis or algorithmic engineering.

IX. CONCLUSION

This paper has outlined the advancement, definition, and theories of BCI in video games, current issues and future advancements. As a result, key terminology and references have been outlined to ensure that researchers and practitioners interested in the field acquire more background information. As proposed elements of the systematic research on BCI, measurement methods, experimental paradigms, pre-processing strategies, and classification methods for the EEG signal were also included.

Discussing the issue within the context of video games, we analyzed the implementation of BCI technology focusing on its ability to change the current approaches to game interfaces. Although there has been an impressive advancement, some problems like signal noise, subject to hardware, as well as the scarcities of resources, are still apparent. But possible future directions and the points of improvement were mentioned, which gave rise to the development of the subject.

As such, this work provides initial insights into the relationship between neuroscience and gaming and points out the challenges that have to be addressed in future research. Thus, by following the directions to innovation mentioned above, BCI technology in future can bring revolutionary changes in gaming and rehabilitation fields and much more.

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